

Project 5: Sudoku Solver - Getting Started

Welcome to your first project. You'll build an AI that solves Sudoku puzzles using Constraint Satisfaction Problems (CSPs).

RESOURCES

- CS50 AI - Week 0: Search https://cs50.harvard.edu/ai/weeks/0/ Focus: Understand DFS, BFS, and backtracking
- CS50 AI - Week 3: Optimization https://cs50.harvard.edu/ai/weeks/3/ Focus: CSP section, constraint propagation, arc consistency

Bonus: Nefreh bikoum if you also complete the course assignments (especially Crossword and Tic-Tac-Toe) - they'll give you solid practice with the concepts.

Tool to explore: Check out pycsp3 or python-constraint library to see how CSPs work in practice. Challenge yourselves to also implement from scratch (even if you end up using a preimplemented solver for the final solution), you can take inspiration from the Crossword assignment code.

IMPLEMENTATION APPROACH

Your main implementation should use backtracking enhanced with CSP techniques (constraint propagation, arc consistency, forward checking).

Optional but recommended: Start with plain DFS/backtracking first, then add CSP improvements. Compare and report the performance difference, you'll learn a lot from seeing the impact of constraint propagation.

2-WEEK SPRINT

Week 1: Study + Build

- Watch lectures and understand core concepts
- Implement your solver (start simple, then add CSP techniques)
- Mid-week check-in: **Wednesday** (ask questions, report progress)

Week 2: Optimize + Polish

- Refine CSP techniques
- Build GUI
- Test on different difficulty levels
- Prepare demo presentation

DELIVERABLES

- **Deadline:** January 1st at 23:59
- **Demo:** TBD

During the demo, you must:

- Present your working Sudoku solver
- Explain DFS, backtracking, and CSP techniques you used
- Show performance comparisons (if you implemented multiple approaches)
- Explain the main characteristics of CSPs

Don’t forget to have fun!

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